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SHORT ITEMS

The "p-hacking-is-terrific" ocean - A cartoon for teaching statistics

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Abstract

P-hacking is fishing for statistical significance through repeated testing on massive data. It would lead to spurious findings, misguide social practice and policy making, and thus should be avoided. Teaching about p-hacking is important, yet challenging. Cartoons are effective edutainment tools to engage students in learning statistical concepts. We created a cartoon and discussed how to use it in teaching about p-hacking by guiding students to think and answer a list of questions. This cartoon can be helpful with teaching both statistics courses and applied seminar courses in various other disciplines. Students are expected to gain a better understanding of multiple issues related to p-hacking, including its occurrence due to repeated testing, the problems with using an arbitrary threshold for the *P*-value and comparing statistical significance, the distinction between statistical vs scientific significance, the approach for interpreting testing results with a holistic view, and the strategies to avoid p-hacking.

KEYWORDS
teaching, cartoon, p-hacking, *P*-value, statistical significance, teaching statistics

1 | INTRODUCTION

The *P*-value, commonly used in inferential statistics and hypothesis testing, is the probability of observing a statistical estimate of the data that is equal to or more extreme than its actually observed value, when the null hypothesis is true. The smaller the *P*-value, the more incompatibility of the data with the null hypothesis, and the greater evidence against the null hypothesis [1]. A widely adopted rule of thumb is to consider $P < .05$ as small enough and providing sufficient evidence to reject the null hypothesis and accept the alternative hypothesis which is usually of interests [1–3]. In this situation, we conclude that the test on the hypotheses is statistically significant. For example, we evaluate a null hypothesis as there being no relationship between a student's love for cartoons and their statistical ability by testing the correlation between measures on these two variables. In the data from a sample of students, if a test of the correlation coefficient yields a *P*-value less than .05, we would reject the null hypothesis and conclude that a student's statistical ability is related to their love for cartoons.

Because the alternative hypothesis often indicates meaningful findings interested by academic journals and funding agencies, researchers are eager for it to be acceptable. *P-hacking* is motivated to search for evidence to

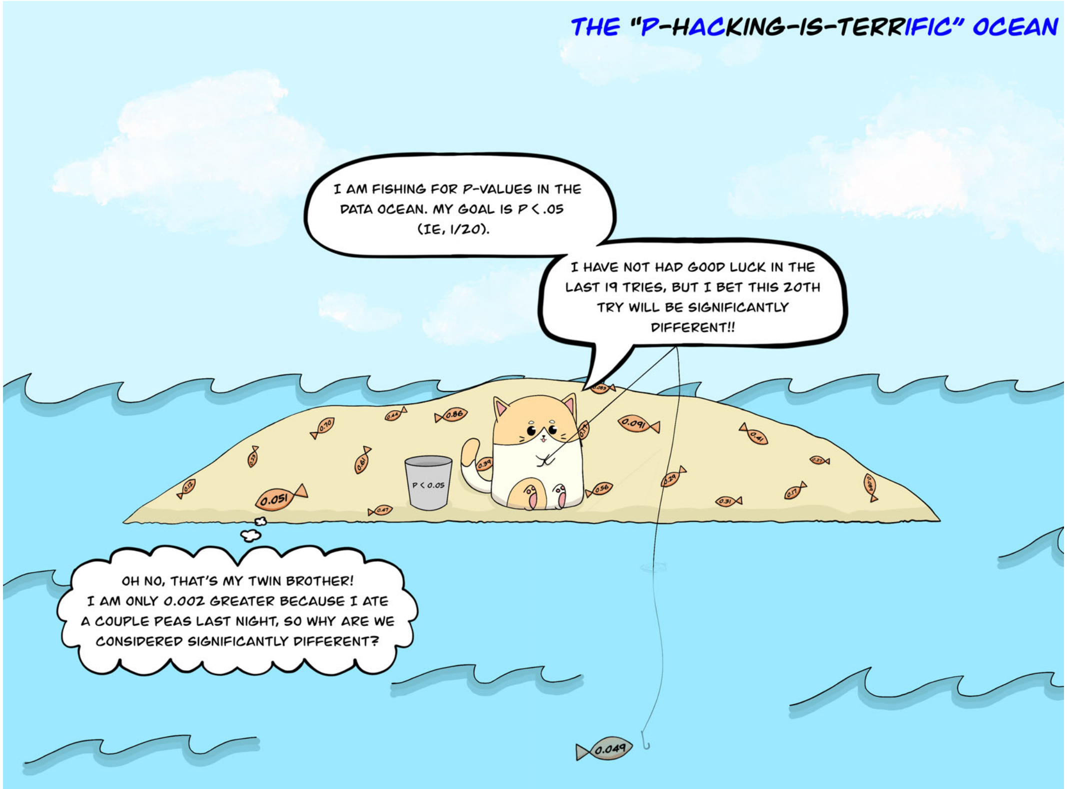

FIGURE 1 The "p-hacking-is-terrific" ocean - a cartoon for teaching statistics

support the alternative hypothesis. It is "the relentless search for statistical significance – comes in many guises. The most pervasive form is also referred to as 'data dredging', 'significance chasing', 'data snooping', or 'going on a fishing expedition', and refers to combining through masses of data, looking for significance, most often without any hypotheses guiding the search" [4, p. 286]. The most common practices of p-hacking include: (a) testing on a vast number of outcomes and cherry picking the ones that are statistically significant; (b) creating new samples by adding more observations or eliminating observations whose data do not support the hypothesis; and (c) trying different statistical testing methods or models until a significant *P*-value is found [4–8]. All these practices would artificially enlarge the data pool for the *P*-value to be fished. As a result, p-hacking would lead to spurious findings that cannot be reproduced in future research, which would in turn misguide social practice and policy making, and thus should be avoided [1,9].

As a serious and complex issue in scientific research, p-hacking is an important topic of statistical education in preparation for future researchers and scholars. However, teaching about p-hacking can be challenging, given that it requires students to first have a good understanding of hypothesis testing and inferential statistics, which are already complex and difficult concepts themselves [10–12]. Research has shown that cartoons are effective edutainment tools to engage students in learning and understanding statistical concepts [13–19]. With this in mind, we created a cartoon about p-hacking and discussed in the next section how it can be used to facilitate teaching.

2 | TEACHING WITH THE CARTOON

As depicted in this cartoon (Figure 1), a cat was fishing for *P*-values in the data ocean. The data ocean represented a pool of all the data created during the p-hacking process, which, as discussed previously, could result from testing multiple outcomes, generating new samples, and trying different statistical methods and models. Each fish represented a statistical estimate from sample testing. The cat targeted to catch a fish with $P < .05$, but had not achieved this goal in the last 19 tries. However, it bet to catch a significantly different fish on the 20th try. Eventually on the 20th try, the cat caught a fish with $P = .049$. However, a previously-caught fish with $P = .051$ commented that they were fish twins and questioned as to why they were considered significantly different.

This cartoon can be presented as an individual or group assignment with several questions for students to think and discuss. Now, students can bring their thoughts and ideas to have a class discussion together. The instructor can guide and summarize the class discussion, as well as provide further answers and advice, to help students better understand the issue. Here, we present a list of example questions that can be asked to students and corresponding answers. We expect that students and teachers may have other questions, answers, and thoughts in viewing this cartoon. For the first three questions, we consider the situation that the null hypothesis is true.

1. Why did the cat bet that the 20th try would be significantly different?
   Each fish represented a statistical estimate from sample testing. All fish formed the distribution of all the estimates. Under the situation that the null hypothesis is true, 5% of the estimates would be extreme enough - that is, deviated far enough from the population value - to yield $P < .05$. The cat assumed that the 20 fish it caught appropriately represented all fish in the data ocean and thus 5% (ie, 1/20) of them would be extreme enough to yield $P < .05$. Given that the first 19 fish all had $P > .05$, the cat bet that the 20th fish would be the extreme one it was looking for. However, a group of 20 fish was a very small sample of estimates and might have not adequately represented all the fish in the data ocean. Therefore, more or less than 5% of them might have actually been extreme enough to yield $P < .05$. That was why the cat was *betting*, but not for sure.
2. What was the probability that the 20th caught fish had $P < .05$?
   Here, we consider the number of fish in the data ocean to be infinite, analogous to the infinite number of estimates obtained from sample testing. Given that catching each fish was an independent event, the probability that the caught fish had $P < .05$ in any try would be the same and equal to 5%. Thus, the probability that the 20th caught fish had $P < .05$ was also 5%.
3. What was the probability that the cat caught at least one fish with $P < .05$ in 5 tries, 10 tries, 20 tries, and 100 tries? What trend of the probability as the number of tries increases do you observe?
   As explained in Question 2, the probability that the caught fish had $P < .05$ in any try was 0.05. Thus, the probability that the caught fish did not have $P < .05$ in any try was $1\text{-}0.05 = 0.95$. Therefore, the probability that the cat did not catch any fish with $P < .05$ in k tries was 0.95^k, and the probability that it caught at least one fish with $P < .05$ in k tries was then $(1\text{-}0.95^k)$, where $k = 1, 2, 3, ..., n$. Correspondingly, the calculated probability for at least one fish with $P < .05$ in 5, 10, 20, and 100 tries is 22.6%, 40.1%, 64.2%, and 99.4%, respectively. As the number of tries increases, this probability goes up, and is almost equal to 100% in 100 tries. As the Nobel Prize-winning economist Ronald Coase [20, p. 16] once said, "If you torture the data long enough, nature will always confess."
4. Why did the fish with $P = .051$ complain? Do you think the two fish $P = .051$ and $P = .049$ were significantly different? Why or why not?
   The fish with $P = .051$ was different in size (analogous to effect size) from all the other caught fish with $P > .05$. However, the cat did not consider them as significantly different, because none of them reached the $P < .05$ cutoff. Compared to the fish with $P = .051$, the fish with $P = .049$ was identical in size. However, only because its *P*-value just passed the $P < .05$ cutoff, the cat considered it as significantly different from all the other caught fish - including the one with $P = .051$. This was why the fish with $P = .051$ complained. Its complaint illustrated two issues in statistical testing. First, the $P < .05$ or any other threshold is an arbitrary dichotomization of the *P*-value continuum. Only a trivial change like eating a couple of peas can move the estimate from one side of the threshold to the other [1,21]. Drawing conclusions solely based on such arbitrary thresholds can lead to the detection of a statistical difference that is not scientifically significant or the omission of a scientific difference that is not statistically significant. Second, whether or not the two fish are significantly different should not be based on the comparison of their *P*-values. One should examine the statistical significance of their difference rather than the difference between their statistical significance. "The difference between 'significant' and 'not significant' is not itself statistically significant." [21, p. 328]. We would not be able to know whether the two fish are significantly different from a statistical perspective until we perform a statistical test on their difference. However, even if the statistical test on their difference was significant, we would doubt that the two fish are significantly different from a scientific perspective given their identical size.
5. What advice would you like to give the cat?
   Although the *P*-value and its statistical significance are important, they are not the only important pieces of information to look at. "God loves the .06 nearly as much as the .05" [22, p. 1277]. Decision making solely based upon the arbitrary dichotomization of statistical significance - for example, $P < .05$ - can lead to incorrect conclusions such as the significance of a trivial

effect or the insignificance of a non-trivial effect. Statistical significance is not equivalent to scientific significance [1,2]. In addition to effect size which represents scientific significance, other factors such as sample size, measurement precision, and types of statistical tests also play a role in statistical significance. One should take a holistic and integrative view of these multiple factors and interpret the *P*-value in context for evaluating the testing results [1,2]. To avoid p-hacking, statistical testing should be guided by well-formed research hypotheses and conducted under a carefully designed study protocol including plans for the number of observations, outcomes to measure, and statistical analysis methods [4,6,9]. Furthermore, other statistical approaches can be used to supplement *P*-values, such as effect size measures, multiple testing corrections, confidence intervals, and Bayesian methods [1,9,23,24].

3 | FINAL REMARKS

This cartoon can be used for teaching both undergraduate and graduatelevel statistics courses that cover hypothesis testing and inferential statistics. It can also be included in seminar courses for research and professional trainees in various other disciplines that talk about study design and applied statistical analysis. Through viewing this cartoon and discussing the questions, students are expected to gain a deeper understanding of multiple issues related to p-hacking. These issues include its occurrence due to repeated testing, the problems with using an arbitrary threshold for the *P*-value and comparing statistical significance, the distinction between statistical vs scientific significance, the approach for interpreting testing results with a holistic view, and the strategies to avoid p-hacking. With a better understanding of these issues, researchers can be more vigilant and thoughtful with their research design, data analysis, and results interpretation, as well as in evaluating the previous findings published in the literature, leaving no space for p-hacking to grow and feel terrific.

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